Root locus technique

Root locus → it is a ploting of system dynamic characteristic Equation.

Ex: consider a system with unity feedback given

$G(s) = \dfrac{K}{s(s+2)}$ $H(s) = 1$

$1 + G(s)H(s) = 0 \Rightarrow 1 + \dfrac{K}{s(s+2)} = 0$

$s^2 + 2s + K = 0$.

$\dfrac{C(s)}{R(s)} = \dfrac{Num}{Den}$

if $K = 0$,

$s^2 + 2s = 0$

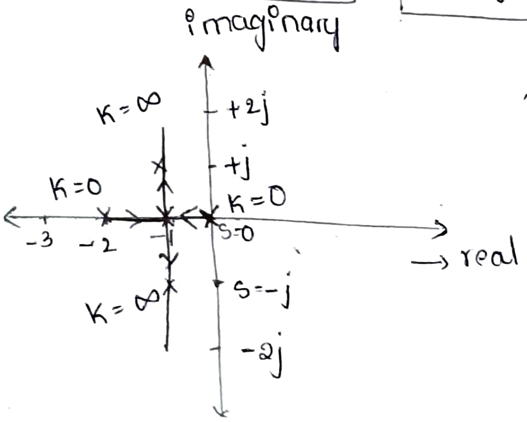
$s(s+2) = 0$ $\boxed{s=0}$, $\boxed{s=-2}$

if $K = 1$.

$s^2 + 2s + 1 = 0$ $\boxed{s=-1}$, $\boxed{s=-1}$.

if $K = 2$

$s^2 + 2s + 2 = 0 \Rightarrow \boxed{s=-1+j}$, $\boxed{s=-1+j}$



root locus exist from 0 to ∞.

★ imp

Rules for construction of Root locus

Rule 1: The root locus is always symmetric about the real axis.

Rule 2: Each branch of root locus always originates from an open loop pole and terminates at either an open loop '0' or '∞'

Rule 2 : The number of branches terminating at ∞ when number of poles greater than number of 0's $(P-Z)$

The number of branches originating at '∞' when number of 0's greater than no. of poles vica versa $(Z-P)$

Rule 3 : The total no of root locii is equals to $\max(p, z)$

Rule 4 : Segments of the real axis having an odd no. of real axis open loop (poles + zeros) to the right or the parts of root locus.

NOTE : For odd number of poles to the right the root locus exist for even of poles root locus doesn't exist.

Rule 5 : The root locus branches that tends to ∞, do so along straight line asymptotes.

- The no of Asymptotes is given by 'A' $= P-Z$
- Angle of asymptotes is given by $\theta A = \dfrac{\pm 180^\circ (2q+1)}{P-Z}$

where q = no of asymptotes. for $2 = 0, 1$
$3 = 0, 1, 2$
$4 = 0, 1, 2, 3$

Rule 6 : The point of intersection of the asymptotes with the real axis is called the centroid. and is given by centroid is equals to

$$\frac{(\Sigma \text{ real part of the poles} - \Sigma \text{ real parts of zeros})}{P-Z}$$

$$C = \frac{\Sigma R_p - \Sigma R_z}{P-Z}$$

Rule 7 : The breakaway point is calculated by solving

$$\frac{dK}{ds} = 0$$

Rule 8 : The angle of departure from an open loop poles is given by $\phi_d = 180^\circ \pm \phi$ where ϕ is the net contribution to the pole P by all other Poles and zeros

Angle of departure is found for complex poles

Rule 9 : The angle of arrival at an open loop zero is given by $\phi_a = 180° + \phi$ complex zero

Rule 10: The intersection root locus branches with the imaginary axis is determined by RH criterion

Problems

1) The open loop TF of a control system is given by $G(s)H(s) = \frac{K}{s(s+1)(s+2)}$ sketch the complete root locus.

To locate Poles & Zeros

Soln Step 1 : $1 + G(s)H(s) = \frac{K}{s(s+1)(s+2)}$

no of zeros $= 0$.

no of poles $= 3$, $s=0, s=-1, s=-2$

Step 2 : no of asymptotes $= P-Z \Rightarrow 3$
$3-0$

Step 3 : Angle of asymptotes. $\frac{\pm 180°(2q+1)}{P-Z}$

$q = 0, 1, 2.$

at $q=0$, $\theta_1 = \frac{\pm 180°(1)}{3} = 60°$,

at $q=1$, $\theta_2 = \frac{\pm 180°(2+1)}{3} = 180°$

at $q=2$ $\theta_3 = \frac{\pm 180°(5)}{3} = 300.$

Step 4 : Centroid of asymptotes $s=0,-1,-2$

$\text{centroid} = \frac{\Sigma R_p - \Sigma R_z}{P-Z} \Rightarrow \frac{0-1-2}{3} \Rightarrow -1$

Step 5 : To find breakaway point

$1 + G(s)H(s) = 0$

$s(s+1)(s+2) + K = 0$

$(s^2+s)(s+2) + K = 0$

$\cancel{s^3 + 2s^2} +$ $s^3 + 3s^2 + 2s + K = 0.$

$K = -s^3 - 3s^2 - 2s$

$\frac{dK}{ds} = -3s^2 - 6s - 2$

$$-3s^2 - 6s - 2 = 0$$

$$3s^2 + 6s + 2 = 0$$

~~$s = -0.31,\ s = 0.151$~~

$\boxed{s = -0.4226}$ $\boxed{s = -1.5773}$

~~$K = s^3 + 3s^2$~~ for $\boxed{s = -0.4226}$

$$K = -s^3 - 3s^2 - 2s \Rightarrow$$

$$\boxed{K = 0.3849}$$

for $\boxed{s = -1.5773}$. × not valid (got −ve) K so,

$$K = -(-1.5773)^3 - 3(-1.5773)^2 - 2(-1.5773)$$

$$\boxed{K = -0.3849}$$

Since, for $s = -1.5773$, K is -0.3849, so it is not a valid breakaway point

$s = -0.4226$ is valid breakaway point.

Step 6: Intersection to imaginary axis

characteristic Eqn $s^3 + 3s^2 + 2s + K = 0$

s^3	1	2
s^2	3	K
s^1	$\frac{6-K}{3}$	0
s^0	K	0.

$\frac{6-K}{3} = 0$, $6 - K = 0$, $\boxed{K_{mar} = 6}$

~~$3s^2 + 6s + 2 = 0$~~

$3s^2 + K_{mar} = 0.$

$3s^2 + 6 = 0.$

$\omega = \pm\sqrt{2}$

$\boxed{s = \pm\sqrt{2}j = \omega}$

Step 7: Plot graph.

Date

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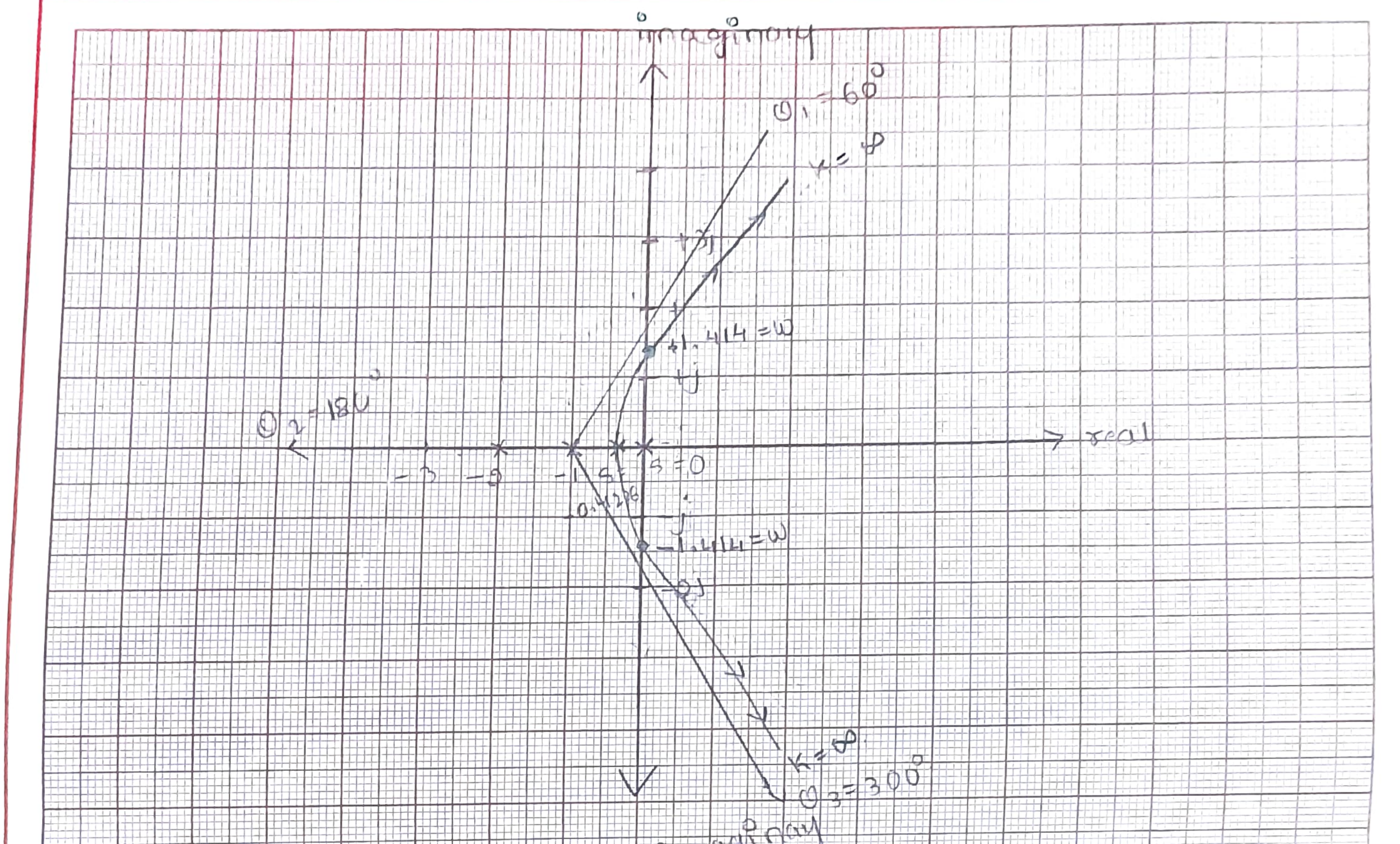


2) Sketch the root locus plot for all values of K ranging from 0 to ∞ for a -ve feedback CS characterised by $G(s)H(s) = \frac{K(s+6)}{s(s+1)(s+2)}$

Step 1: no of zeros $\rightarrow$ 1 at -6

no of poles $\rightarrow$ 3 at $s = 0, -1, -2$

Step 2: no of asymptotes $= P - Z \Rightarrow 3 - 1 \Rightarrow 2$.

Step 3: Angle of asymptotes $= \pm \frac{180^\circ(2q+1)}{P-Z}$

$q = 0, \quad \frac{180^\circ}{2} \Rightarrow 90^\circ$

$q = 1, \quad \frac{180^\circ(3)}{2} = 270^\circ$

14/5/2021

Human welfare

Introduction to value Education.

Centroid $\Rightarrow \dfrac{\Sigma R_p - \Sigma R_z}{P-Z} \Rightarrow \dfrac{-3+6}{2} \Rightarrow \dfrac{3}{2} \Rightarrow 1.5$

To find breakaway point

$1+G(S)H(S) = S(S+1)(S+2) + SK + 6K$

$S^3+3S^2+2S+SK+6K = 0 \Rightarrow S^3+3S^2+(2+K)S+6K$

$S^3+3S^2+2S+(S+6)K = 0.$

$K = \dfrac{-S^3-3S^2-2S}{S+6}.$

$$\frac{dK}{dS} = \frac{(S+6)(-3S^2-6S-2)-(-S^3-3S^2-2S)1}{(S+6)^2}$$

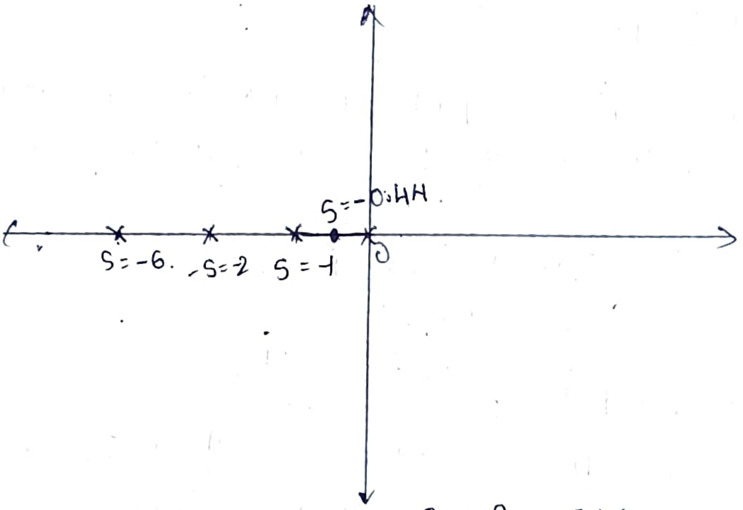
$$= \frac{-3S^3-6S^2-2S-18S^2-36S-12+S^3+3S^2+2S}{S^2+36+12S}$$

$$= \frac{-2S^3-21S^2-36S-12}{S^2+36+12S} = 0.$$

$-2S^3-21S^2-36S-12 = 0$

$2S^3+21S^2+36S+12 = 0$

$\boxed{S=-8.45}$ ✗, $\boxed{S=-0.44}$ ✓, $\boxed{S=-1.6}$ ✗



The break away point is -0.44.

$$s^3 + 3s^2 + 2s + (s+6)K = 0 \qquad s^3 + 3s^2 + (2+K)s + 6K$$

s^3	1	$2+K$
s^2	3	$6K$
s^1	$\frac{6-3K}{3}$	0
s^0	$6K$	$0.$

$$\frac{3(2+K) - 6K}{3} \qquad \frac{6+3K-6K}{3}$$

$$\frac{6-3K}{3}$$

$$\frac{6-3K}{3} = 0$$

$$6 - 3K = 0, \qquad \Rightarrow 6 = 3K$$

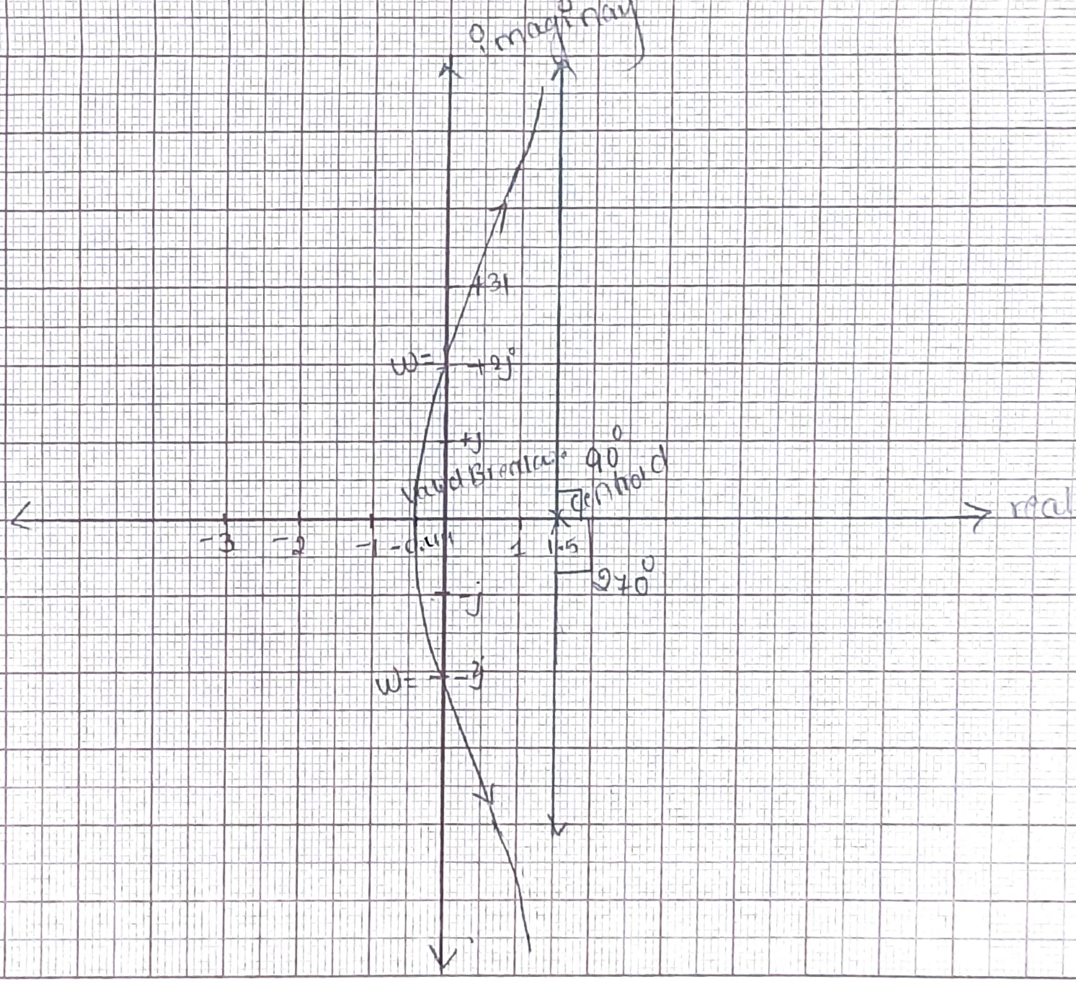
$$\boxed{K_{mar} = 2}$$

$$3s^2 + 6K = 0.$$

$$3s^2 + 6(2) = 0 \quad \Rightarrow 3s^2 + 12 = 0 \quad \Rightarrow 3s^2 = -12$$

$$s^2 = \frac{-\cancel{12}\,4}{\cancel{3}} \quad \Rightarrow s = \pm 2$$

$$\boxed{\omega = \pm 2 \text{ rad/sec}}$$



imaginary
real
+3j
ω=
+2j
+j
90°
centroid
-3
-2
-1
-0.44
1
1.5
270°
-j
ω=

2. Sketch the root locus diagram for the unity feedback CS with $G(s) = \frac{K}{s(s^2+8s+17)}$ determine the value of K for a damping ratio of 0.5

$$1+G(s)H(s) = \frac{K}{s(s^2+8s+17)} \Rightarrow s^3+8s^2+17s=0$$

no of zeros = 0.

no of poles = 3, at $s=0$, $s=-4+j$, $s=-4-j$

No. of Asymptotes = $P-Z \Rightarrow 3-0 = 3$.

$q = 0, 1, 2$.

at $q=0$, $\theta_1 = \frac{\pm 180^\circ}{3} \Rightarrow 60^\circ$.

$q=1$, $\theta_2 = 180^\circ$, $q=2$, $\theta_3 = 300^\circ$.

centroid of asymptotes

real part

$$\text{Centroid} = \frac{\Sigma Rp - \Sigma Rz}{P-Z} \Rightarrow \frac{-4-4}{3} \Rightarrow -2.667$$

step 5: To find breakaway point

$1+G(s)H(s)=0$

$$1 + \frac{K}{s^3+8s^2+17s} = 0$$

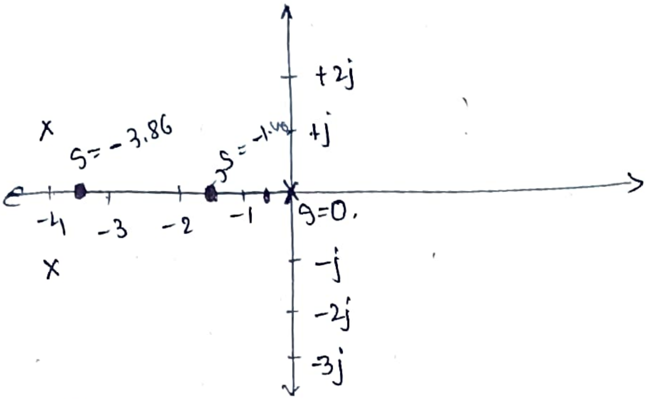
(8) $s^3+8s^2+17s+K=0.$

$K=-s^3-8s^2-17s$

$\frac{dK}{ds} = -3s^2-16s-17$

$-3s^2-16s-17=0$ $\quad$ $3s^2+16s+17=0$

$\boxed{s=-1.46}$, $\boxed{s=-3.86}$.



$s^3+8s^2+17s+K=0.$

s^3	1	17
s^2	8	K
s^1	$\frac{136-K}{8}$	0
s^0	K	0.

$\frac{136-K}{8}=0$

$136-K=0$

$\boxed{K_{max}=136}$

$8s^2+K=0$

$8s^2+136=0.$

$8s^2=-136$ $\quad$ $\boxed{s=\pm 4.12j}$

$\boxed{\omega = \pm 4.12 \text{ rad/sec}}$

$\theta_d = 180^\circ-(165.96+90^\circ)$

$\theta_d=-75.96$ at $-4+j$

$+75.96$ at $-4-j$

(4) To find K value given $\xi=0.5$

Let $\alpha=\cos^{-1}\xi$ $\therefore$ $\alpha=\cos^{-1}0.5$ $\quad$ $\boxed{\alpha=60^\circ}$

$|G(s).H(s)|/_{at\ s=s_d}=1$

$s=-4+j$

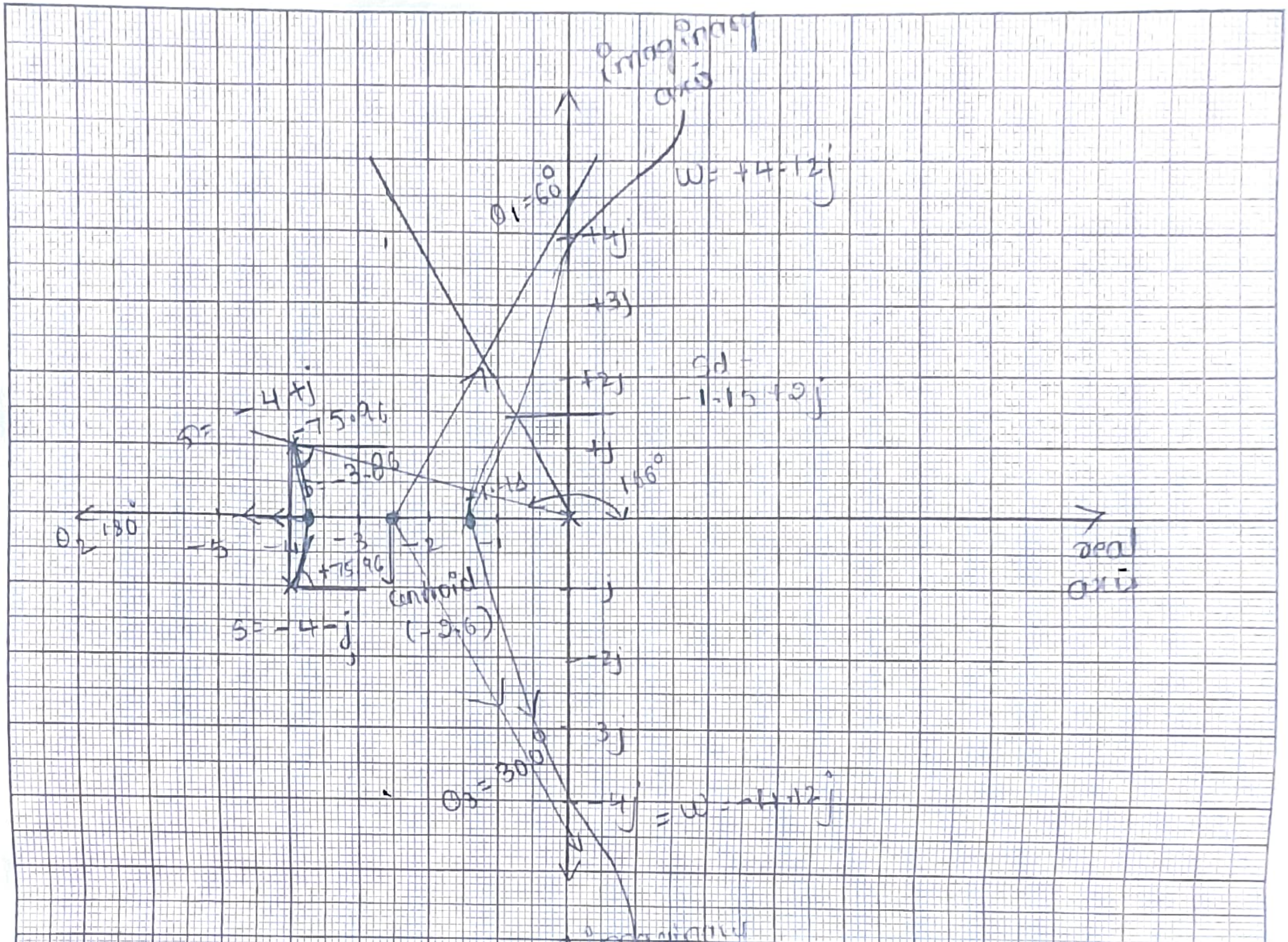
$$G_1(s) = \left| \frac{K}{s(s^2+8s+17)} \right| = 1 \quad s = -1.15+2j$$

$$= \frac{K}{-1.15+2j\left[(-1.15+2j)^2+8(-1.15+2j)+17\right]} = 1$$

$$= \frac{K}{(-1.15+2j)\left[\,5.12+11.4j\,\right]} = 1$$

$$= \frac{K}{\sqrt{(-1.15)^2+2^2}\,\sqrt{(5.12)^2+(11.4)^2}} = 1$$

$$\boxed{K = 28.6}$$


imaginary axis
ω = +4.12j
θ1 = 60°
+4j
+3j
+2j
+j
sd = -1.15 + 2j
s = -4 + j
-75.96
-3.86
1.13
156°
θ2 180°
-5
-4
-3
-2
-1
real axis
+75.96
centroid
(-2.6)
s = -4 - j
-j
-2j
-3j
θ3 = 300°
-4j = ω = -4.12j

④ sketch the root locus diagram for open loop TF

$$G(s)\ H(s) = \frac{K}{s(s+2)(s+5)}$$

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5) The open loop TF of a control s/m is given by $G(s) = \frac{K}{s(s+1)(s+2)(s+3)}$ sketch the complete root locus. Find the critical value of K (Kmax) and location of roots on the imaginary axis.

soln

$$1 + G(s)\ H(s) = \frac{K}{s(s+1)(s+2)(s+3)}$$

no of zeros = 0.

no of poles = 4, $s=0$, $s=-1$, $s=-2$, $s=-3$

no of asymptotes = $P-Z \Rightarrow 4-0 \Rightarrow 4$

Angle of asymptotes $\frac{\pm 180^\circ(2q+1)}{P-Z}$.

$q = 0, 1, 2, 3.$

$\boxed{q=0,\ \theta_1 = 45^\circ}$ $\boxed{q=1,\ \theta_2 = 135^\circ}$

$\boxed{q=2,\ \theta_3 = 225^\circ}$ $\boxed{q=3,\ \theta_4 = 315^\circ}$

centroid of asymptotes

$$\text{Centriod} = \frac{\Sigma R_p - \Sigma R_z}{P - Z} \Rightarrow \frac{0-1-2-3}{4} \Rightarrow -1.5$$

$1 + G(s)H(s) = 0$

$s(s+1)(s+2)(s+3) + K = 0$

$(s^2+s)(s^2+3s+2s+6) + K = 0$

$s^4 + 3s^3 + 2s^3 + 6s^2 + s^3 + 3s^2 + 2s^2 + 6s + K = 0$

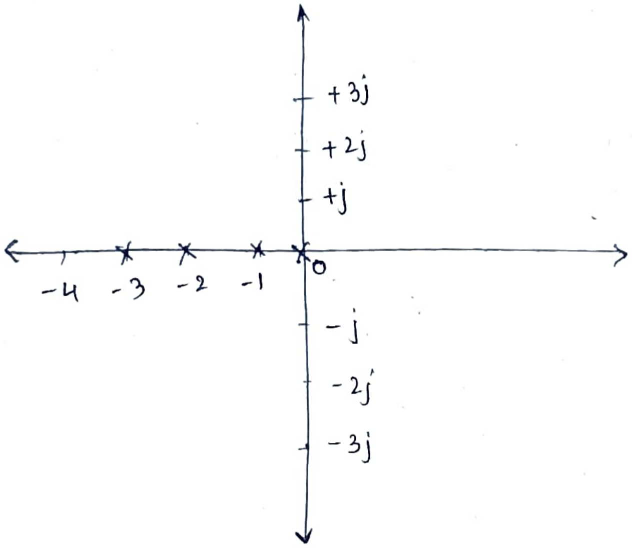
$s^4 + 6s^3 + 11s^2 + 6s + K = 0$

$K = -s^4 - 6s^3 - 11s^2 - 6s$

$\frac{dK}{ds} = -4s^3 - 18s^2 - 22s - 6.$

$4s^3 + 18s^2 + 22s + 6 = 0$

$\boxed{s = -0.38}$ $\boxed{s = -2.61}$ $\boxed{s = -1.5}$



$s^4 + 6s^3 + 11s^2 + 6s + K = 0$

s^4	1	11	K
s^3	6	6	0
s^2	10	K	0
s^1	$\frac{60-6K}{10}$	0	0
s^0	K	0	0

$\frac{60-6K}{10} = 0$

$60 - 6K = 0$

$60 = 6K$

$\boxed{K = 10}$

$\boxed{K_{max} = 10}$

$$10s^2 + 15 = 0$$

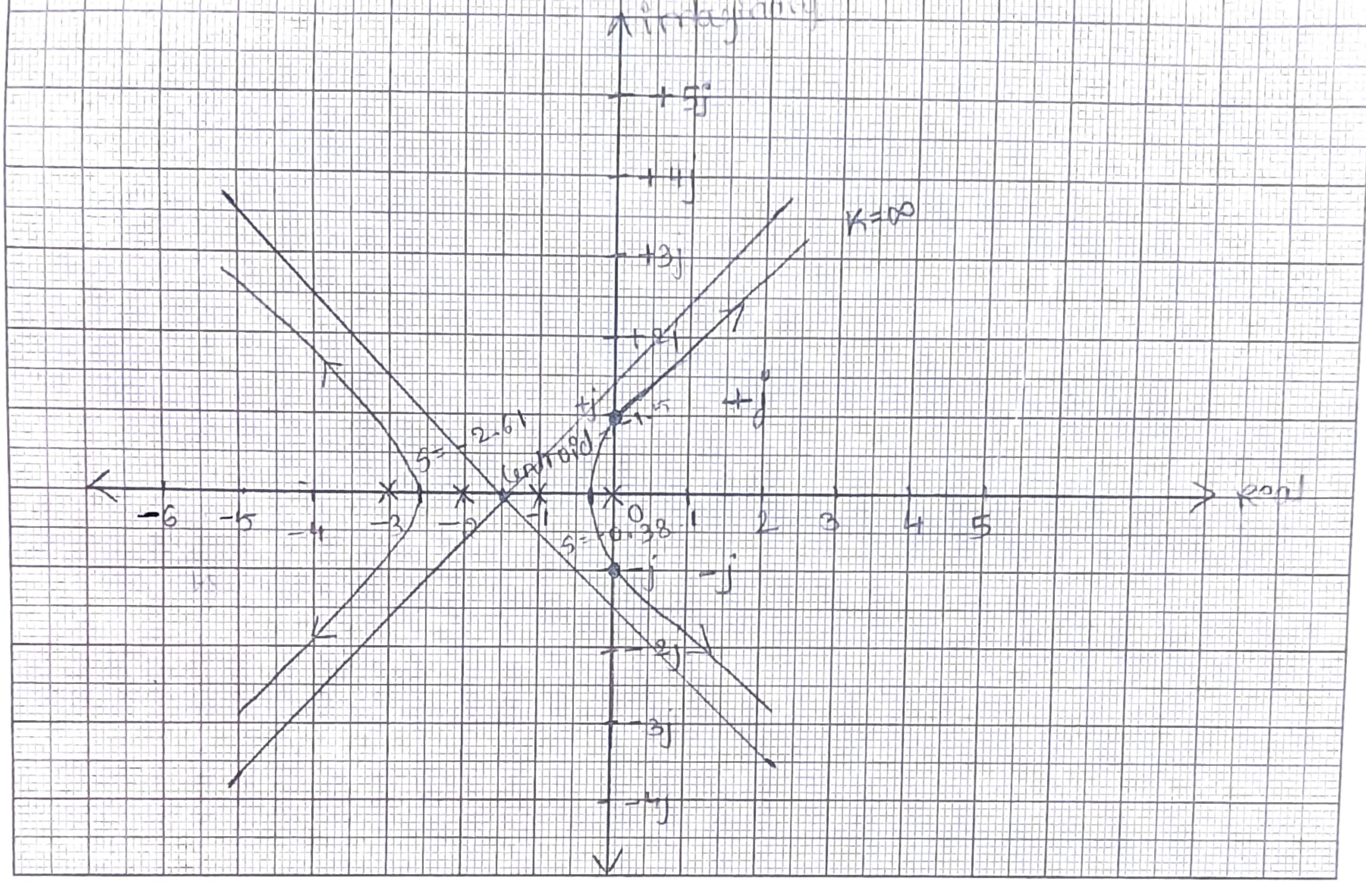
$$10s^2 + 10 = 0$$

$$10s^2 = -10$$

$$s^2 = -1$$

$$\boxed{s = \pm j}$$

$$\boxed{\omega = \pm 1 \text{ rad/sec}}$$



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6 Prove that part of root locus is a circle using angle condition and find the center as well as radius when $G(s)H(s) = \frac{K(s+2)}{s(s+1)}$

Soln Given $G(s)H(s) = \frac{K(s+2)}{s(s+1)}$ replace s by ~~$(\alpha+j\omega)$~~ $(\alpha+j\beta)$

$$\cancel{G(j\omega)H(j\omega) = \frac{K(j\omega+2)}{j\omega(j\omega+1)}}$$

$$G(\alpha+j\beta)\,H(\alpha+j\beta) = \frac{K(\alpha+j\beta+2)}{(\alpha+j\beta)(\alpha+j\beta+1)}$$

$$\angle G(s)H(s) = \frac{\tan^{-1}\left(\frac{0}{K}\right) + \tan^{-1}\left(\frac{\beta}{\alpha+2}\right)}{\tan^{-1}\left(\frac{\beta}{\alpha}\right) + \tan^{-1}\left(\frac{\beta}{\alpha+1}\right)}$$

WKT, $\tan^{-1}A \pm \tan^{-1}B \Rightarrow \tan^{-1}\left(\frac{A \pm B}{1 \mp AB}\right)$

$$= \frac{\tan^{-1}\left[\dfrac{\frac{0}{K} + \frac{\beta}{\alpha+2}}{1 - \left(\frac{0}{K}\right)\left(\frac{\beta}{\alpha+2}\right)}\right]}{\tan^{-1}\left[\dfrac{\frac{\beta}{\alpha} + \frac{\beta}{\alpha+1}}{1 - \left(\frac{\beta}{\alpha}\right)\left(\frac{\beta}{\alpha+1}\right)}\right]}$$

$$= \frac{\tan^{-1}\left(\frac{\beta}{\alpha+2}\right)}{\tan^{-1}\left[\frac{\beta/\alpha + \beta/\alpha+1}{1-(\beta/\alpha)(\beta/\alpha+1)}\right]}$$

$$= \tan^{-1}\left(\frac{\beta}{\alpha+2}\right) - \tan^{-1}\left[\frac{\frac{\beta(\alpha+1)+\alpha\beta}{\alpha(\cancel{\alpha+1})}}{\frac{\alpha(\alpha+1)-\beta^2}{\alpha(\cancel{\alpha+1})}}\right]$$

$$= \tan^{-1}\left(\frac{\beta}{\alpha+2}\right) - \tan^{-1}\left[\frac{\alpha\beta+\beta+\alpha\beta}{\alpha^2+\alpha-\beta^2}\right]$$

$$= \tan^{-1}\left(\frac{\beta}{\alpha+2}\right) - \tan^{-1}\left[\frac{2\alpha\beta+\beta}{\alpha^2+\alpha-\beta^2}\right]$$

$$= \frac{\tan^{-1}\left(\frac{\beta}{\alpha+2} - \frac{2\alpha\beta+\beta}{\alpha^2+\alpha-\beta^2}\right)}{1+\left(\frac{\beta}{\alpha+2}\right)\left(\frac{2\alpha\beta+\beta}{\alpha^2+\alpha-\beta^2}\right)}$$

= According to the angle condition

$\angle G(s)H(s) = 180^\circ$ for a point on the root locus.

$\therefore \tan^{-1}(180^\circ) = 0.$

$$\frac{\tan^{-1}\left(\frac{\beta}{\alpha+2} - \frac{2\alpha\beta+\beta}{\alpha^2+\alpha-\beta^2}\right)}{1+\left(\frac{\beta}{\alpha+2}\right)\left(\frac{2\alpha\beta+\beta}{\alpha^2+\alpha-\beta^2}\right)} = 0$$

$$\frac{\beta}{\alpha+2} - \frac{2\alpha\beta+\beta}{\alpha^2+\alpha-\beta^2} = 0.$$

$$\frac{\beta}{\alpha+2} = \frac{2\alpha\beta+\beta}{\alpha^2+\alpha-\beta^2}$$

$$(\alpha^2+\alpha-\beta^2)\beta = (\alpha+2)(2\alpha\beta+\beta)$$

$$\alpha^2\beta + \alpha\beta - \beta^3 = 2\alpha^2\beta + \alpha\beta + 4\alpha\beta + 2\beta$$

$\beta[\alpha^2+\alpha-\beta^2] = \beta[-4\alpha+2+2\alpha^2+\alpha]$

$\alpha^2+\cancel{\alpha}-\beta^2 - 4\alpha - 2\alpha^2 - \cancel{\alpha} - 2 = 0$

$-\alpha^2-4\alpha-\beta^2-2=0 \Rightarrow \alpha^2+4\alpha+\beta^2+2=0.$

Add and subtract $+2$ and -2

$\therefore \alpha^2+4\alpha+\beta^2+2+2-2=0.$

$\alpha^2+4\alpha+\beta^2+4 = 2$

$\alpha^2+4+4\alpha+\beta^2 = 2 \Rightarrow (\alpha+2)^2+\beta^2 = (\sqrt{2})^2 \longrightarrow (1)$

Equation of circle is given by

$\boxed{(x-h)^2+(y-k)^2 = r^2} \longrightarrow (2)$

where, center $=(h,k)$ radius $=r$.

$\therefore$ Equation (1) resembles eqn of a circle

$\therefore$ The part of root locus is a circle with center $(-2, 0)$ and radius $\sqrt{2} \Rightarrow 1.41$.

(4) Show that the part of the root locus for open loop TF $G(s)H(s) = \dfrac{K(s+3)}{s(s+2)}$ is a circle. Also find the radius and center of the circle.

$C = -3, 0.$
$r = \sqrt{3}.$

Soln $G(s)H(s) = \dfrac{K(s+3)}{s(s+2)}$

replace s by $(\alpha+j\beta)$

$G(s)H(s) = \dfrac{K[\alpha+j\beta+3]}{(\alpha+j\beta)[\alpha+j\beta+2]}$

$\underline{|G(s)H(s)} =$

Bode plot

① A unity feedback system as $G(s) = \dfrac{20}{s(s+2)(s+20)}$ draw the Bode plot. Determine gain margin gm, Pm, wgc, wpc, and also comment on the stability.

Sol Step 1: Arrange G(s) H(s) in time constant form

$$G(s)\,H(s) = \frac{\cancel{20}\ \cancel{40}\ 2}{(\cancel{20})\,s\,(\cancel{2})[1+s/2][1+s/20]}$$

$$G(s)H(s) = \frac{2}{s(1+s/2)(1+s/20)} \Rightarrow \frac{2}{s(1+0.5s)(1+0.05s)}$$

(den)

$\boxed{\text{NOTE}}$: if 1 pole at the origin

(s) slope = −20dB/dec

(s^2) = −40 dB/dec

(s^3) = − 60 dB/dec

(num)

if 1 zero at the origin

(s) slope = ±20 dB/dec

s^2 = ±40 dB/dec

s^3 = ±60 dB/dec.

Step 2: Identify the factors

(i) K = 2

(ii) one pole is at origin = −20dB/dec

(iii) one simple pole,

$$\frac{1}{1+0.5s} \quad \text{with } T_1 = 0.5$$

$$\therefore W_{c1} = \frac{1}{T_{c1}} = \frac{1}{0.5} = 2 \text{ rad/sec}$$

(iv) One more simple pole

$$\frac{1}{1+0.05s} \quad \text{with } T_2 = 0.05$$

$$\therefore W_{c2} = \frac{1}{T_2} = \frac{1}{0.05} = 20 \text{ rad/sec.}$$

Step 3: Magnitude plot analysis.

for reference line, $20\log K = 20\log 2$

$= 6\,dB$

terms

terms	corner frequency (ascending order)	slope	change in slope
$\frac{1}{s}$	—	-20dB/dec	-20dB/dec
$\frac{1}{1+0.5s}$	2 rad/sec	-20dB/dec	$\xrightarrow{+}$ -40dB/dec
$\frac{1}{1+0.05s}$	20 rad/sec	-20dB/dec	$\xrightarrow{+}$ -60dB/dec

Step 4: phase plot analysis.

replace s by $j\omega$

NOTE: In the phase plot analysis for poles take $-\tan^{-1}$ for zeros $+\tan^{-1}$

$$\frac{1}{s} = \frac{1}{j\omega} \Rightarrow -\tan^{-1}\left(\frac{\omega}{0}\right) \Rightarrow -\tan^{-1}(\infty) = \underline{\underline{-90^\circ}}$$

$$\frac{1}{1+0.5s} = \frac{1}{1+0.5j\omega} \Rightarrow -\tan^{-1}\left(\frac{0.5\omega}{1}\right) \Rightarrow \underline{\underline{-\tan^{-1}(0.5\omega)}}$$

$$\frac{1}{1+0.05s} = \frac{1}{1+0.05j\omega} \Rightarrow -\tan^{-1}\left(\frac{0.05\omega}{1}\right) \Rightarrow \underline{\underline{-\tan^{-1}(0.05\omega)}}$$

ω_c = 0.2 2 5 10 20 100 ∞.

ω_c	-90°	$-\tan^{-1}(0.5\omega)$	$-\tan^{-1}(0.05\omega)$	ϕ_R
0.2	-90°	-5.71°	-0.57°	-96.28
2	-90°	-45°	-5.71°	-140.71°
5	-90°	-68.19°	-14.03°	-172.22°
10	-90°	-78.69°	-26.56°	-195.25°
20	-90°	-84.28°	-45°	-219.28°
100	-90°	-88.85°	-78.69°	-257.54°
∞	-90°	-90°	-90°	-270°

The intersection of magnitude plot with 0dB line ~~Blane~~ line is the gain cross over freq

The intersection of phase plot with -180° line phase cross over freq

$\omega gc = 2$ rad/sec

$\omega pc = 6.3$ rad/sec

if $\omega pc > \omega g$

$\frac{1}{1+ST}$

Phase margin $= 38°$.

Gain margin $= 22$ dB.

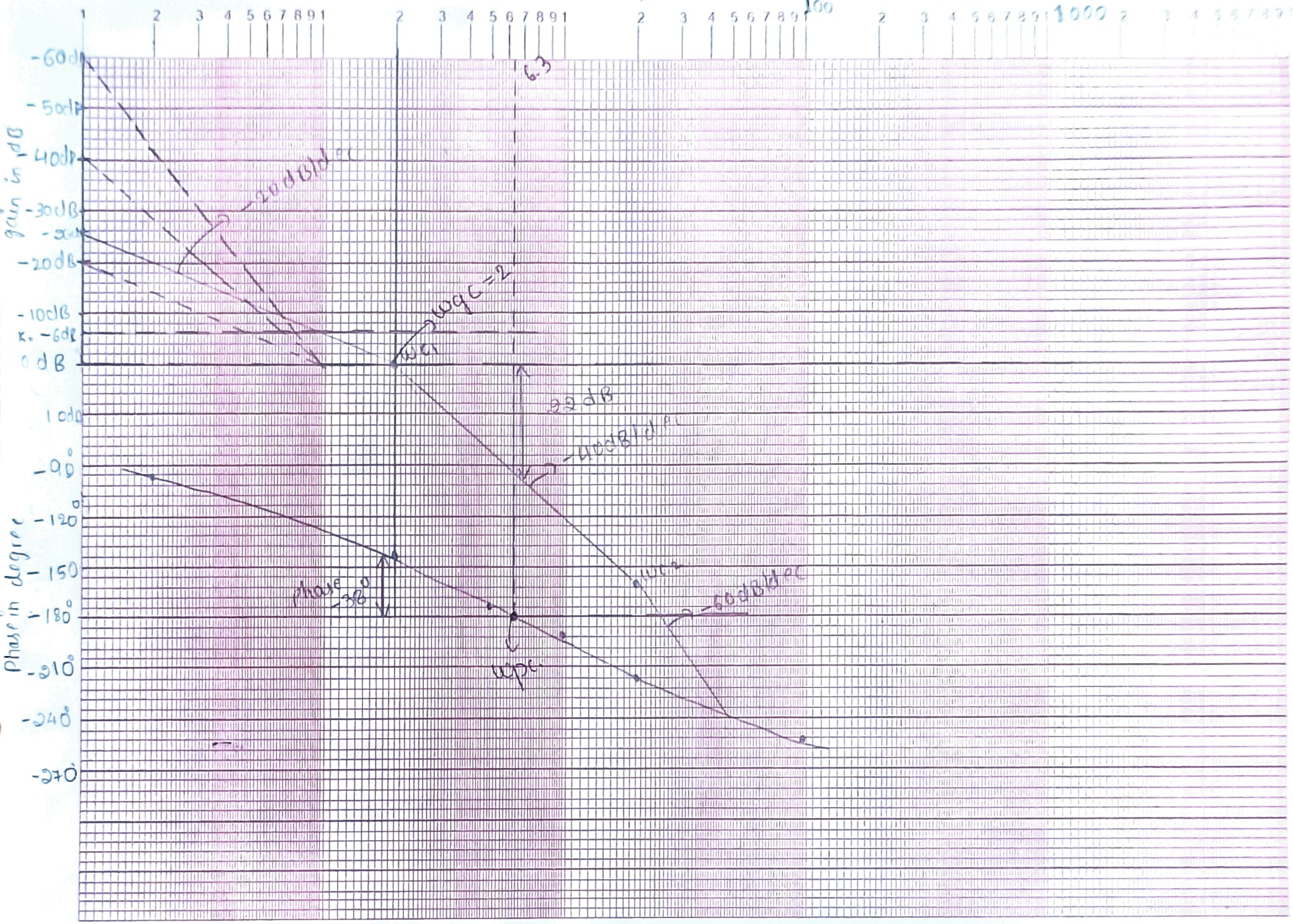
Hence, the system is stable,

if $\omega gc > \omega pc$ → s/m is unstable.

if $\omega pc = \omega gc$ → s/m is marginary stable.

① magnitude

1 division = 20dB (magnitude plot)
1 division = 60dB (Phase)



log(ω) ⟶

1 div = 60°

Q)* For the Slm having open loop TF given by

$$G(S)H(S) = \frac{10(1+0.125S)}{S(1+0.5S)(1+0.25S)}$$ draw the Bode plot

determine Pm, gm, wgc, wpc and also comment for the stability

Soln Arrange G(S) H(S) in time constant form

$$G(S)H(S) = \frac{10(1+0.125S)}{S(1+0.5S)(1+0.25S)}$$

(i) $K = 10$,

(ii) one pole at origin $= \frac{1}{S} =$ -20dB/dec

(iii) one simple pole, $\frac{1}{1+0.5S}$ $T_1 = 0.5$ $\omega_{c1} = 2$ rad/sec

one simple pole, $\frac{1}{1+0.25}$ $T_2 = 0.25$ $\omega_{c2} = 4$ rad/sec

one simple zero, $(1+0.125S)$ $T_3 = 0.125$ $\omega_{c3} = 8$ rad/sec

magnitude plot analysis

$20\log 10 \Rightarrow 20$ dB.

Terms	Corner frequency (Ascendening)	Slope	change in Slope
$\frac{1}{s}$	—	$-20dB/dec$	$-20dB/dec$
$\frac{1}{1+0.5s}$	2 rad/sec	$-20dB/dec$	$-40dB/dec$
$\frac{1}{1+0.25s}$	4 rad/sec	$-20dB/dec$	$-60dB/dec$
$1+0.125s$	8 rad/sec	$+20dB/dec$	$-40dB/dec$

Phase plot analysis

replace s by $j\omega$.

$\frac{1}{s} = \frac{1}{j\omega} \Rightarrow -\tan^{-1}\left(\frac{\omega}{0}\right) \Rightarrow -\tan^{-1}(\infty) = -90^{\circ}$

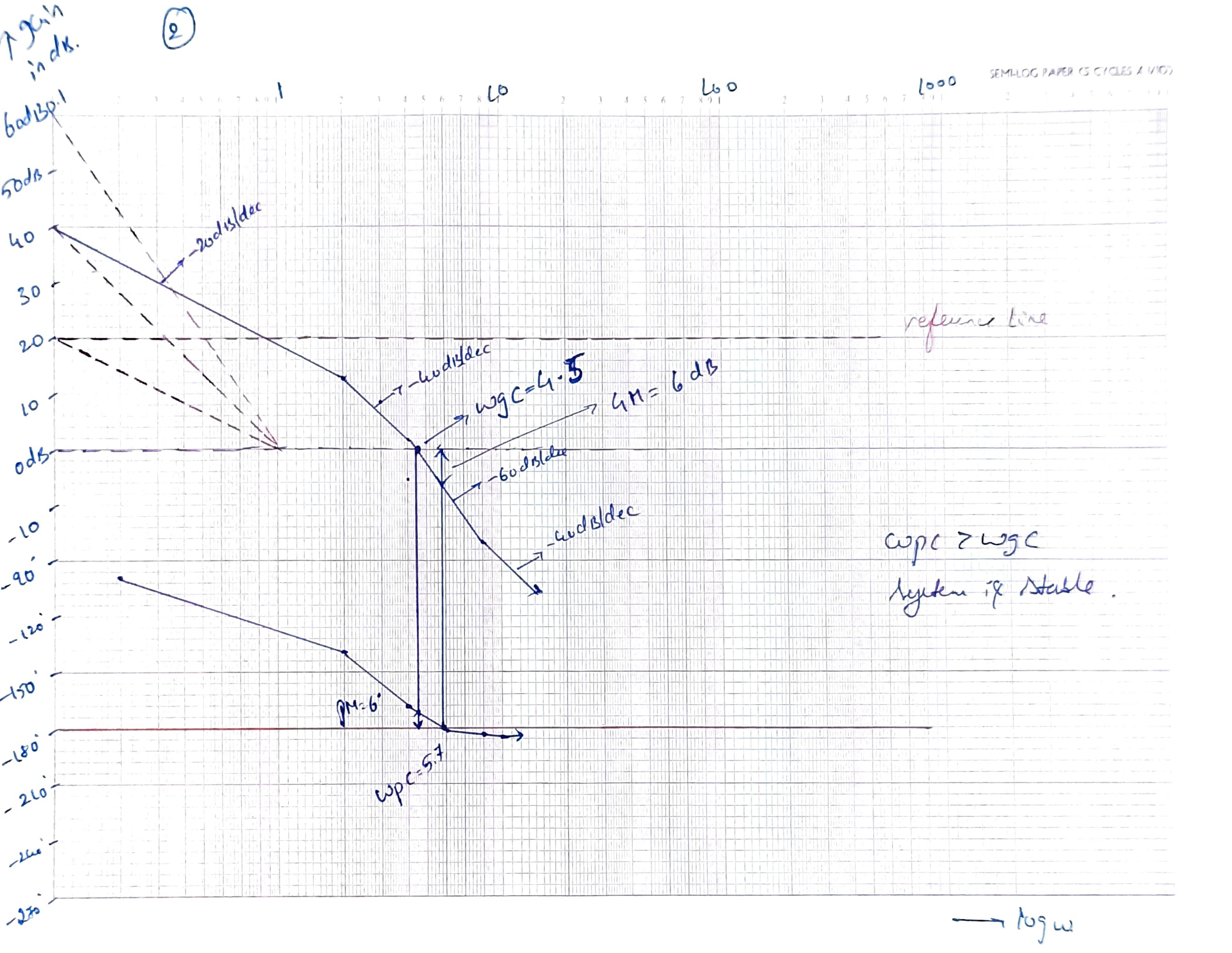
$\frac{1}{1+0.5s} \Rightarrow \frac{1}{1+0.5j\omega} \rightarrow -\tan^{-1}\left(\frac{0.5\omega}{1}\right) \Rightarrow -\tan^{-1}(0.5\omega)$

$\frac{1}{1+0.25s} \Rightarrow \frac{1}{1+0.25j\omega} \Rightarrow -\tan^{-1}\left(\frac{0.25\omega}{1}\right) \Rightarrow -\tan^{-1}(0.25\omega)$

$1+0.125s \Rightarrow 1+0.125j\omega \Rightarrow \tan^{-1}(0.125\omega)$

$\omega_c \rightarrow$ 0.2 2 4 8 10 ∞

ω_c	-90°	$-\tan^{-1}(0.5\omega)$	$-\tan^{-1}(0.25\omega)$	$\tan^{-1}(0.125\omega)$	ϕ_R
0.2	-90°	-5.71°	-2.86°	$+1.43$	-97.14°
2	-90°	-45°	-26.56°	$+14.03$	-147.53°
4	-90°	-63.43°	-45°	$+26.5^{\circ}$	-171.93°
8	-90°	-75.96°	-63.43°	45°	-184.39°
10	-90°	-78.69°	-68.19°	51.34°	-185.54
∞	-90°	-90°	-90°	90°	-180°



↑ gain in dB.
2
SEMI-LOG PAPER (5 CYCLES X 1/10)
0.1
1
10
100
1000
60dB
50dB
40
30
20
10
0dB
-10
-90°
-120°
-150°
-180°
-210°
-240°
-270°
-20dB/dec
reference line
-40dB/dec
ωgc=4.5
GM = 6 dB
-60dB/dec
-40dB/dec
ωpc > ωgc
System is Stable.
PM=6°
ωpc=5.7
→ log ω

2) Plot the Bode diagram for the Open loop TF

$$G(s)\,H(s) = \frac{10}{S(1+0.4S)(1+0.1S)}$$

Step 1: Arrange G(s) H(s) in time constant form

$$G(s)\,H(s) = \frac{10}{S(1+0.4S)(1+0.1S)}$$

Step 2: Identify the factors

(i) $K = 10$.

(ii) One pole is at origin = -20dB

(iii) one simple pole,

$\frac{1}{1+0.4S}$ with $T_1 = 0.4$ $\omega_{c1} = \frac{1}{T_1} = \frac{1}{0.4} = 2.5$ rad/sec

(iv) one simple pole,

$\frac{1}{1+0.05S}$ with $T_2 = 0.05 \Rightarrow \omega_{c2} = 20$ rad/sec

Step 3: magnitude plot analysis

for reference line, $20\log K \Rightarrow 20\log 10 \Rightarrow \underline{20dB}$

terms	corner frequency	slope	change in slope
$\frac{1}{s}$	-	-20dB/dec	-20dB/dec
$\frac{1}{1+0.4s}$	2.5 rad/sec	-20dB/dec	-40dB/dec
$\frac{1}{1+0.1s}$	10 rad/sec	-20dB/dec	-60dB/dec

step 4 : Phase plot analysis

replace s by jω

$\frac{1}{s} \Rightarrow \frac{1}{j\omega} \Rightarrow -\tan^{-1}(\omega) \Rightarrow -\tan^{-1}(\infty) \Rightarrow -90^\circ$

$\frac{1}{1+0.4s} \Rightarrow \frac{1}{1+0.4j\omega} \Rightarrow -\tan^{-1}\left(\frac{0.4\omega}{1}\right) \Rightarrow -\tan^{-1}(0.4\omega)$

$\frac{1}{1+0.1s} \Rightarrow \frac{1}{1+0.1j\omega} \Rightarrow -\tan^{-1}\left(\frac{0.1\omega}{1}\right) \Rightarrow -\tan^{-1}(0.1\omega)$

ωc = 0.2 2.5 5 10 20 100

ωc	-90°	$-\tan^{-1}(0.4\omega)$	$-\tan^{-1}(0.1\omega)$	ϕ_R
0.2	-90°	-4.57°	-1.145°	-95.64°
2.5	-90°	-45°	-14.03°	-149.2°
5	-90°	-63.43°	-26.56°	-180°
10	-90°	-75.96°	-45°	-210.96°
20	-90°	-82.87°	-63.43°	-236.3°
100	-90°	-88.56°	-84.28°	-262.8°
∞	-90°	-90°	-90°	-270°

GM = 12 DB

PM = 32°

ω_{pc} = 4.8 rad/sec

ω_{gc} = 2.5 rad/sec

$\omega_{pc} > \omega_{gc}$

∴ The sm is stable.

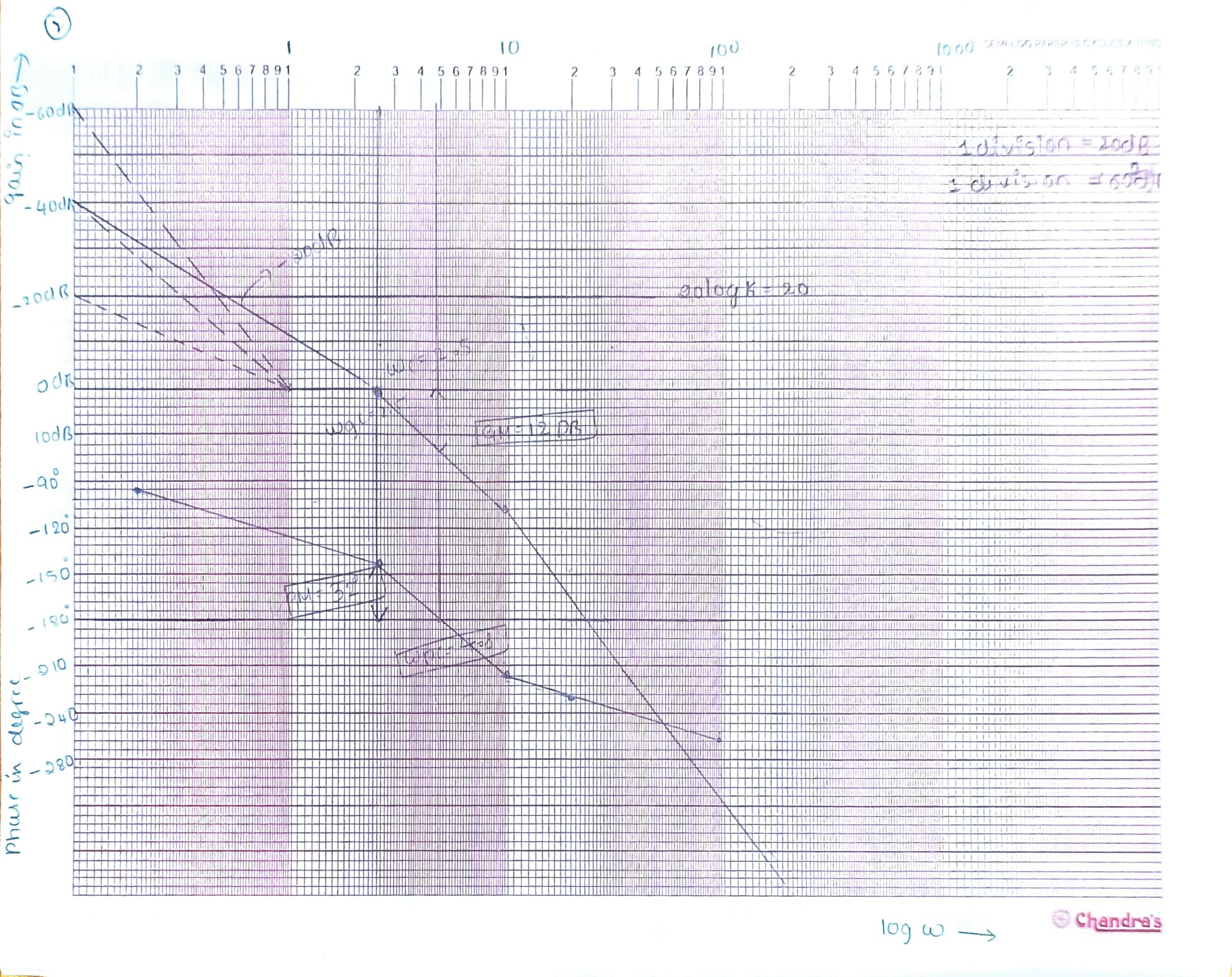
$\omega_{gc} = 2.5$ rad/sec

$\omega_{pc} = 4.8$ rad/sec

Phase margin $= -32^\circ$

Gain margin $= 12$ dB

$\omega_{pc} > \omega_{gc}$ $\therefore$ s/m is stable.



1
10
100
1000
SEMI-LOG PAPER 5 CYCLES
gain in dB
-60dB
-40dB
-20dB
0dB
10dB
-90°
-120°
-150°
-180°
-210
-240
-280
Phase in degree
1 division = 20dB
1 division = 60°
-20dB
20logK = 20
ωc = 2.5
ωg1
GM = 12 DB
PM = 32°
log ω

4) $\frac{5}{s(1+0.5s)(1+0.05s)}$

Step 1: Arrange G(s) H(s) in time constant form

$$G(s)\ H(s) = \frac{5}{s(1+0.5s)(1+0.05s)}$$

Step 2: Identify the factors

(i) $K = 5$

(ii) One pole is at origin = -20dB.

(iii) one simple pole, $\frac{1}{1+0.5s}$ $T_1 = 0.5 \Rightarrow \omega_{c_1} = 2$ rad/sec

another simple pole = $\frac{1}{1+0.05s}$ $T_2 = 0.05$ $\omega_{c_2} = 20$ rad/sec

Step 3: magnitude plot analysis.

for reference. $20\log K$, $20\log 5 \Rightarrow 13.97 \Rightarrow 14$ dB.

terms	Corner freq	Slope	Change in Slope.
$\frac{1}{s}$	-	-20dB/dec	-20dB/dec
$\frac{1}{1+0.5s}$	2 rad/sec	-20dB/dec	-40dB/dec
$\frac{1}{1+0.05s}$	20 rad/sec	-20dB/dec	-60dB/dec

Step 4 : Phase plot analysis

replace s by jω

$\frac{1}{s} \Rightarrow \frac{1}{jω} \Rightarrow -\tan^{-1} ω \Rightarrow -\tan^{-1}(\infty) = \cancel{\tan} -90^\circ$

$\frac{1}{1+0.5s} \Rightarrow \frac{1}{1+0.5jω} \Rightarrow -\tan^{-1}(0.5ω) \Rightarrow -\tan^{-1}(0.5ω)$

$\frac{1}{1+0.05s} \Rightarrow \frac{1}{1+0.05jω} \Rightarrow -\tan^{-1}(0.05ω) \Rightarrow -\tan^{-1}(0.05ω)$

$ω_c$ = 0.2 2 5 10 20 100 ∞.

$ω_c$	-90°	$-\tan^{-1}(0.5ω)$	$-\tan^{-1}(0.05ω)$	ϕ_R
0.2	-90°	-5.71°	-0.57°	-96.28°
2	-90°	-45°	-5.71°	-140.71°
5	-90°	-68.19°	-14.03°	-172.22°
10	-90°	-78.69°	-26.56°	-195.25°
20	-90°	-84.28°	-45°	-219.28°
100	-90°	-88.85°	-78.69°	-257.54°
∞	-90°	-90°	-90°	-270°

$ω_{pc}$ = 6.7 rad/sec

$ω_{gc}$ = 2 rad/sec

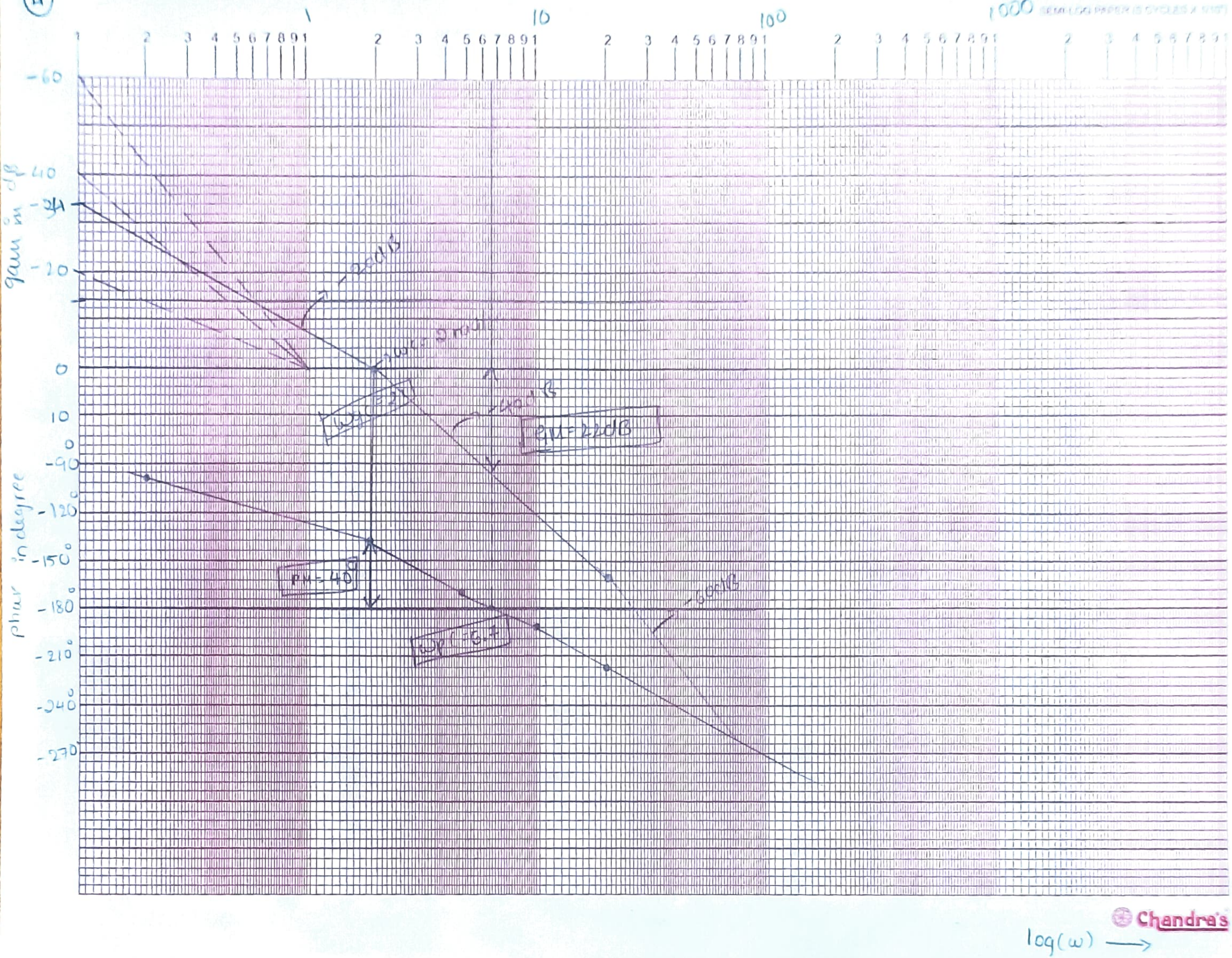
GM = 22 dB

pm = -40°

$ω_{pc} > ω_{gc}$

∴ system is stable.

4



gain in dB
phase in degree
-20dB
-40dB
-60dB
GM = 22dB
PM = 40°
ωpc = 6.7
Chandra's
log(ω)

⑤ Plot the Bode diagram for the Open loop TF of a unity feedback s/m given by

$$G(S) = \frac{100(0.1S+1)}{S(S+1)^2(0.01S+1)}$$

find Gm, Pm, ωgc, ωpc & also comment on the stability of s/m

Sol^n Step 1

G(S) H(S) is in time constant form

Step 2 : identify the factor

(i) $K = 100$

(ii) one pole at origin $\frac{1}{S}$ = −20dB/dec

(iii) one simple pole of second order $\frac{1}{(1+S)^2}$ = −40dB/dec

$T_1 = 1$, $\boxed{\omega_{c_1} = 1 \text{ rad/sec}}$ = ~~20~~dB

(iv) one simple pole $\frac{1}{1+0.01S}$ with $T_2 = 0.01$

$\boxed{\omega_{c_2} = 100 \text{ rad/sec}}$ = −20dB/dec

(v) one simple zero, $(1+0.1S)$ $T_3 = 0.1$ $\boxed{\omega_{c_3} = 10 \text{ rad/s}}$

= +20dB/dec

Step 3 : magnetitude plot analysis.

$20 \log K \Rightarrow 20 \log 100 = 40$ dB.

terms	ωc in Ascending order	slope	change in slope
$\frac{1}{S}$	–	−20dB/dec	−20dB/dec
$\frac{1}{(1+S)^2}$	1 rad/sec	−40 dB/dec	−60 dB/dec
$1+0.1S$	10 rad/sec	+20 dB/dec	−40dB/dec
$\frac{1}{1+0.01S}$	100 rad/sec	−20dB/dec	−60dB/dec

$\omega_c =$ 0.5 1 5 10 50 100 ∞

step 4 : phase plot analysis

~~wc~~ ~~-90°~~

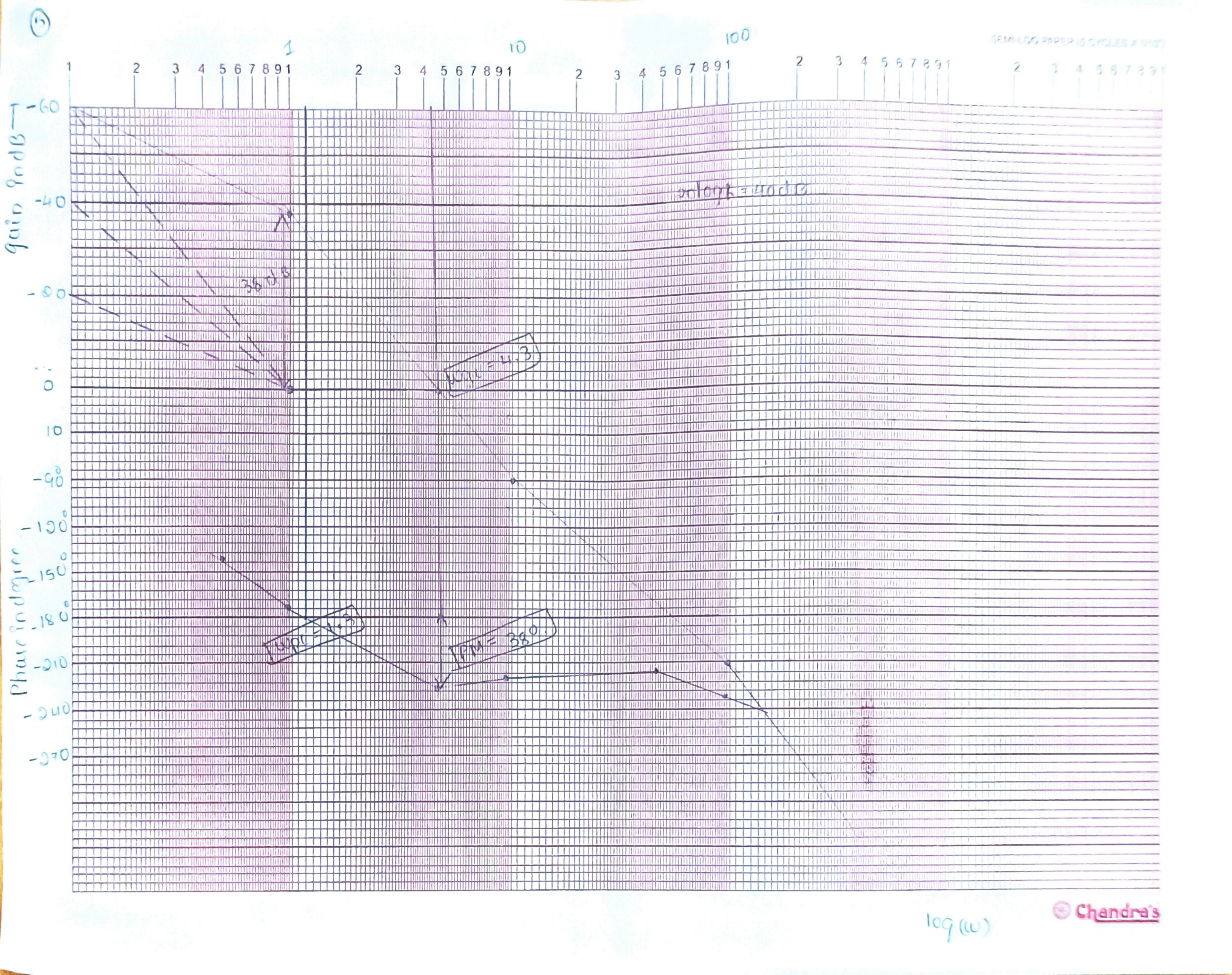
$\frac{1}{s} \Rightarrow \frac{1}{j\omega} \Rightarrow -\tan^{-1}(\omega) \Rightarrow -\tan^{-1}(\infty) = -90°$

$\frac{1}{(1+s)^2} \Rightarrow \frac{1}{(1+j\omega)^2} \Rightarrow -2\tan^{-1}(\omega) \Rightarrow -$

$1+0.1s \Rightarrow 1+0.1j\omega \Rightarrow \tan^{-1}(0.1\omega)$

$\frac{1}{1+0.01s} \Rightarrow \frac{1}{1+0.01j\omega} \Rightarrow -\tan^{-1}(0.01\omega)$

ω_c	$-90°$	$-2\tan^{-1}(\omega)$	$\tan^{-1}(0.1\omega)$	$-\tan^{-1}(0.01\omega)$	ϕ_R
0.5	-90°	-53.13°	2.86°	-0.28°	-140.55°
1	-90°	-90°	5.71°	-0.57°	-174.86°
5	-90°	-157.38°	26.56°	-2.86°	-223.68°
10	-90°	-168.57°	45°	-5.71°	-219.28°
50	-90°	-177.7°	78.69°	-26.56°	-215.57°
100	-90°	-178.8°	84.28°	-45°	-229.52
∞	-90°	-180°	+90°	-90°	-270°



3
SEMI-LOG PAPER (5 CYCLES X 10)
1
10
100
gain in dB
Phase in degree
20log k = 40dB
38dB
Wpc = 4.3
Wgc = 1.3
PM = 38°
-60
-40
-80
0
10
-90
-120°
-150°
-180°
-210
-240
-270
log(w)
Chandra's